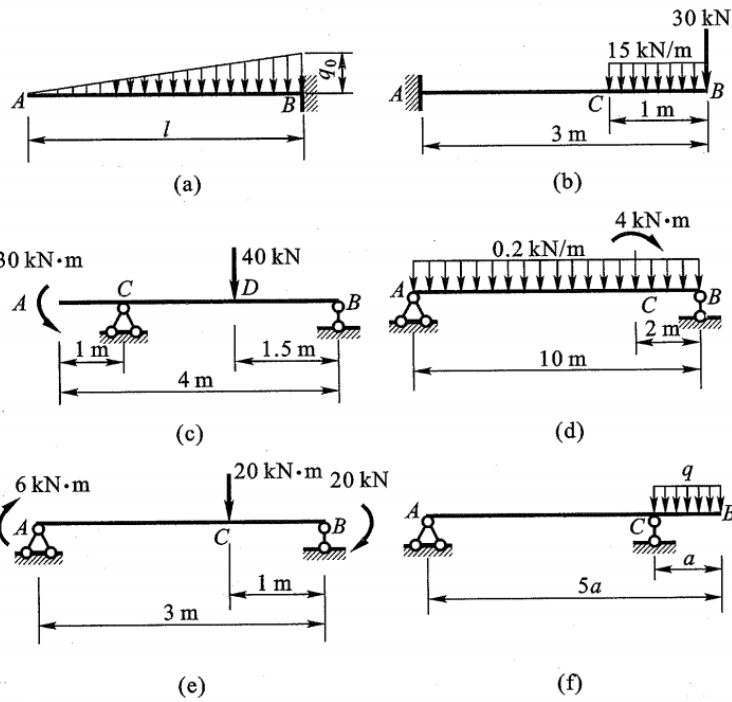


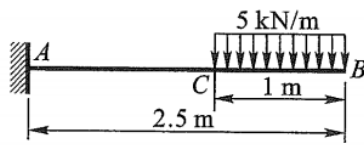
第3次作业——材料力学回顾

材料力学回顾作业及答案

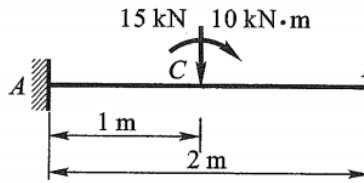
一、画剪力图和弯矩图



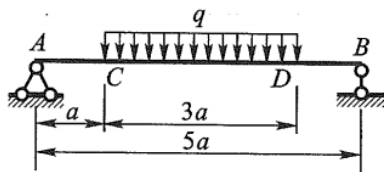
二、试利用荷载集度、剪力和弯矩间的微分关系作下列各梁的剪力图和弯矩图。



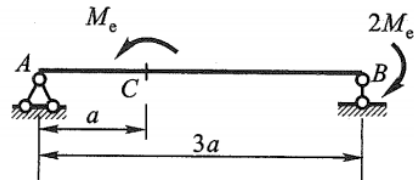
(a)



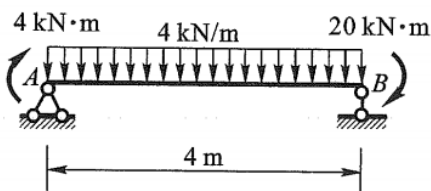
(b)



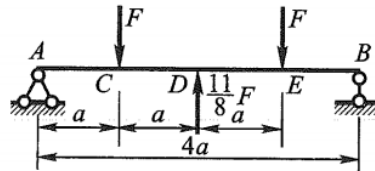
(c)



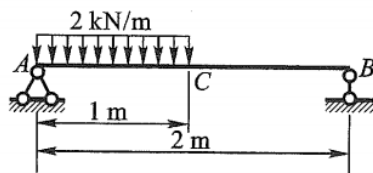
(d)



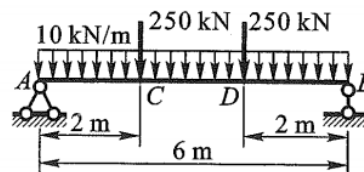
(e)



(f)

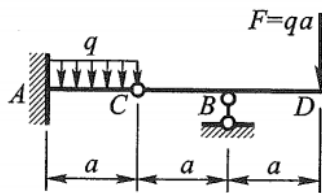


(g)

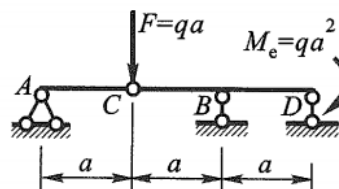


(h)

三、试作下列具有中间铰的梁的剪力图和弯矩图。



(a)

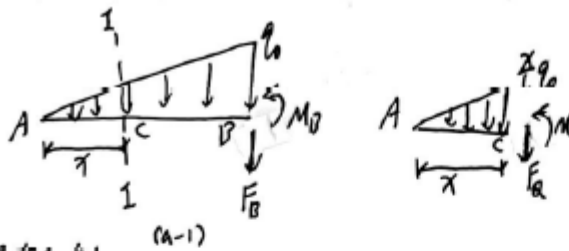


(b)



一、
(a)

梁的受力分析如图(a-1)所示。



可得截面内力

1-1截面上 由 $\sum F_y = 0$, $F_Q = -\frac{1}{2} \cdot x \cdot \frac{x}{l} q_0 = -\frac{q_0}{2l} x^2$, $0 \leq x \leq l$.

$\sum M_C = 0$, $M = -\frac{1}{2} \cdot x \cdot \frac{x}{l} q_0 \cdot \frac{1}{3} \cdot x = -\frac{q_0}{6l} x^3$, $0 \leq x \leq l$.

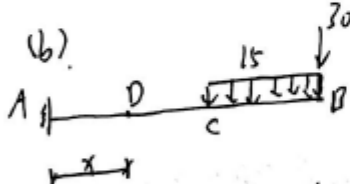
弯矩、剪力图如图(a-2)、(a-3)所示。



Q图
(a-2)

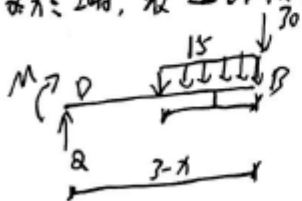


M图
(a-3)



由于梁上受力情况不同,分段讨论。

当 $0 \leq x \leq 2$ 时, 取 1-1 段隔离体

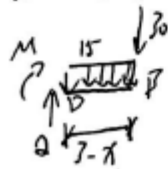


由 $\sum F_y = 0$, $Q = 15x + 30 = 45 \text{ kN}$.

$\sum M_A = 0$, $M + 30 \times (3-x) + 15 \times x \times (\frac{x}{2} - x) = 0$.

$M = -\frac{255}{2} + 45x$

当 $2 \leq x \leq 3$ 时, 取 1-1 段隔离体



由 $\sum F_y = 0$, $Q = 15 \times (3-x) + 30 = 75 - 15x$.

$\sum M_A = 0$, $M + 30 \times (3-x) + 15 \times (3-x) \times \frac{(3-x)}{2} = 0$.

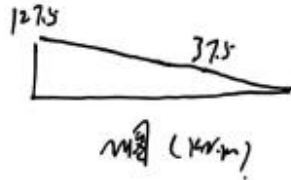
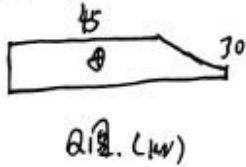
$M = -157.5 + 75x - 7.5x^2$



则1b)图剪力和弯矩方程为

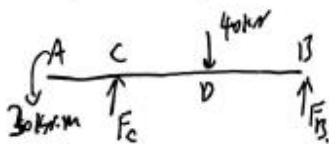
$$Q = \begin{cases} 45 \text{ kN}, & 0 \leq x \leq 2 \\ (75 - 15x) \text{ kN}, & 2 \leq x \leq 3 \end{cases}; \quad M = \begin{cases} (45x - 127.5) \text{ kN}\cdot\text{m}, & 0 \leq x \leq 2 \\ (-157.5 + 75x - 7.5x^2) \text{ kN}\cdot\text{m}, & 2 \leq x \leq 3 \end{cases}$$

弯矩、剪力图如下图所示。



(c).

梁受力如图b所示。



由平衡方程 $\sum M_B = 0$, $F_c \cdot 3 = 30 + 40 \times 1.5$, $F_c = 30 \text{ kN}$.

$\sum F_y = 0$, $F_c + F_b = 40$, $F_b = 10 \text{ kN}$.

取AC段隔离体, 由于AC段剪力为0, 可得 $M_c = 30 \text{ kN}\cdot\text{m}$.

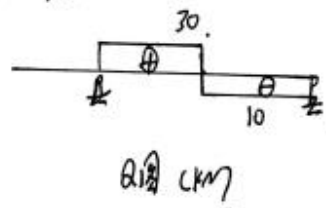
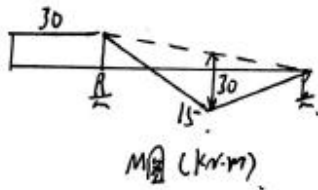
(A—C) M_c
30

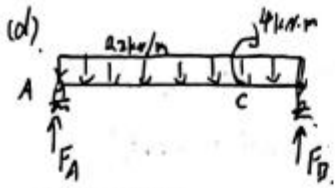
连接B、C点弯矩, 其中 $M_B = 0$ 。在BC段弯矩图上叠加集中力 40 kN 作用在简支梁上的弯矩图。

弯矩叠加 $\frac{1}{4} \times 40 \times 3 = 30 \text{ kN}\cdot\text{m}$ 。

弯矩图如下所示。

由弯矩图斜率可得剪力图。



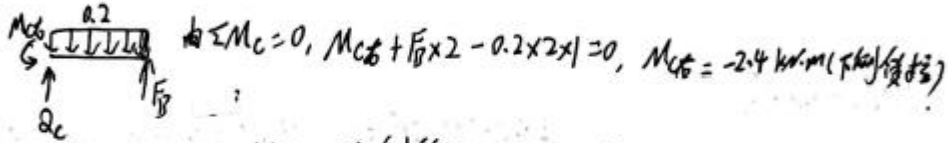


取整体隔离体, 由 $\sum M_A = 0$, $0.2 \times 10 \times 5 + 4 - F_B \times 10 = 0$, 得 $F_B = 1.4 \text{ kN}$.

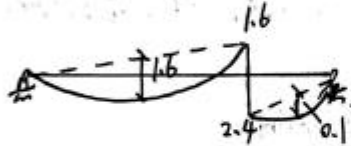
$$\sum F_y = 0, F_A + F_B = 0.2 \times 10, F_A = 0.6 \text{ kN}.$$

C点处作用有集中力偶, 故弯矩图有突变,

取 C 右侧部分隔离体.



由 C 点平衡可得 $M_{C左} = 1.6 \text{ kN}\cdot\text{m}$ (上侧受拉), 则弯矩图为



M 图 (kN·m)

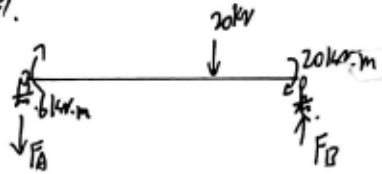
由于梁上无剪力作用, 故剪力图为无突变, 为一条直线, 斜率为 0.2 kN/m .



Q 图 (kN)



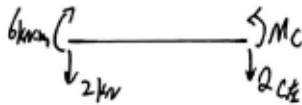
(e).



取整体为隔离体, 对A点取矩, 由 $\sum M_A = 0$, $6 + 20 + 20 \times 2 - F_B \times 3 = 0$, $F_B = 22 \text{ kN}$.

$$\sum F_y = 0, F_A + 20 - F_B = 0, F_A = 2 \text{ kN}.$$

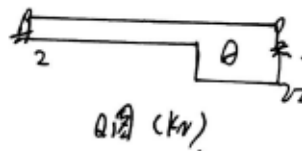
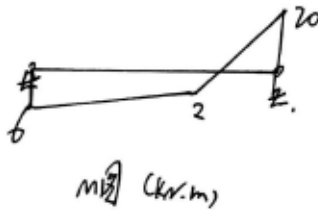
取AC段隔离体



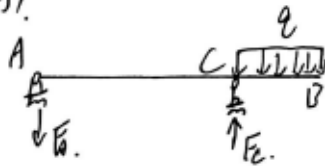
$$\sum M_C = 0, M_C + 2 \times 2 - 6 = 0, M_C = 2 \text{ kN}\cdot\text{m}.$$

$$\sum F_y = 0, Q_{C左} + 2 = 0, Q_{C左} = -2 \text{ kN}.$$

分别连接CA点弯矩和CB点弯矩即为M图, 由M图斜率可得Q图.



(f).



取整体隔离体, 由 $\sum M_A = 0$, $F_B \cdot 4a - q \cdot a \cdot \frac{1}{2}a = 0$, $F_B = \frac{1}{8}qa$.

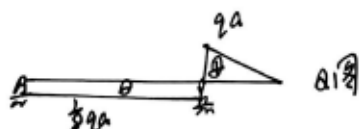
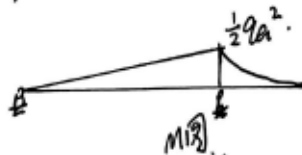
$$\sum F_y = 0, F_A + qa - F_B = 0, F_A = \frac{1}{8}qa.$$

由BC段隔离体,



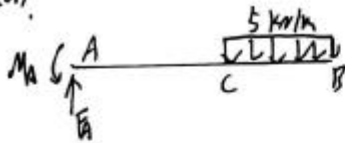
$$\sum M_C = 0, M_C = \frac{1}{2}qa^2.$$

连接A、C点弯矩, 即为M图, 由M图斜率可得Q图.



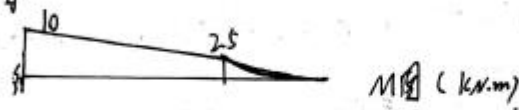
二、

(a)



取整体为隔离体，各段取矩，由 $\sum M_A = 0$ ， $M_A = 5 \times 2 \times 2 = 10 \text{ kN}\cdot\text{m}$
 $\sum F_y = 0$ ， $F_B = 5 \times 2 = 10 \text{ kN}$ 。

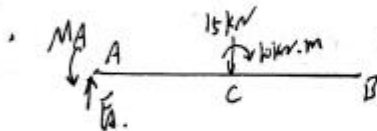
由 AC 段剪力为 5 kN，可得 AC 段弯矩斜率为 5 kN，则 $M_C = 2.5 \text{ kN}\cdot\text{m}$ (上侧受拉)
 则弯矩图为



AC 段剪力不变，为 5 kN，BC 段剪力斜率为均布荷载大小 5 kN/m。

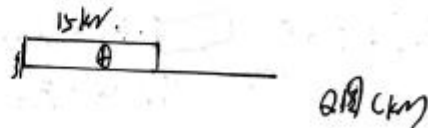
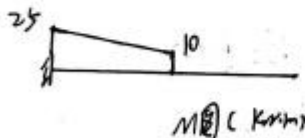


(b)

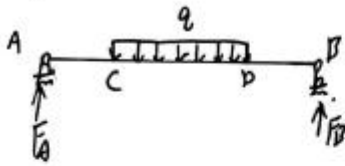


BC 段剪力为 0，则斜率为 0， $Q_{BC} = 0$ 。

取整体隔离体， $\sum M_A = 0$ ， $M_A = 15 \times 1 + 10 = 25 \text{ kN}\cdot\text{m}$ 。
 连接 A、C 弯矩即为 M 图，其余斜率可知剪力图。

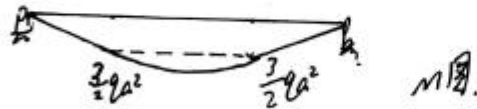


(c).

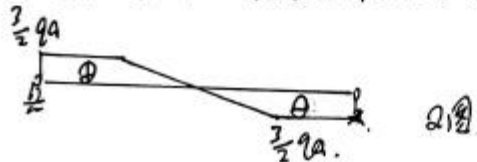


由左右对称可得 $F_A = F_B = \frac{1}{2} \times 3qA = \frac{3}{2}qA$.

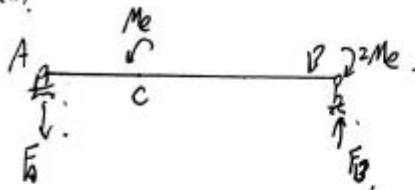
根据 $Q_{AC} = \frac{3}{2}qA$, $M_A = 0$, 可得 $M_{CA} = \frac{3}{2}qA^2$ (下侧受拉) $= M_{DB}$.
连接 C-D 弯矩, 叠加均布荷载作用在简支梁上的弯矩图即为 M 图.



$Q_{AC} = \frac{3}{2}qA$, $Q_{DB} = -\frac{3}{2}qA$, CD 段剪力斜率 k 为 q , 则



(d).

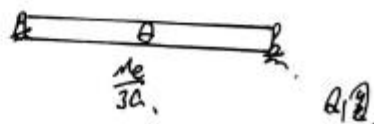
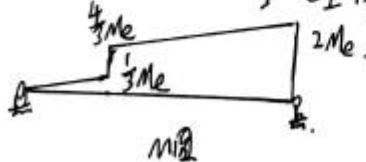


取整体隔离体, $\sum M_A = 0$, $2Me - Me - F_B \cdot 3A = 0$, $F_B = \frac{Me}{3A}$.

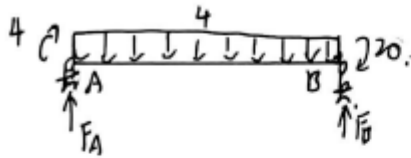
$$\sum F_y = 0 \quad F_A = F_B = \frac{Me}{3A}$$

由 AC 段剪力为 $Q_{AC} = -\frac{Me}{3A}$, $M_A = 0$, 可得 $M_{CA} = \frac{Me}{3}$ (上侧受拉).

根据 C 点平衡, 得 $M_{CB} = \frac{4Me}{3}$ (上侧受拉), 则 M 图为



1e).



取整体隔离体, 由 $\sum M_B = 0$, $4 \times 4 \times 2 + 20 - F_A \times 4 = 0$, $F_A = 14 \text{ kN}$.

$$\sum F_y = 0, F_A + F_B = 4 \times 4, F_B = 2 \text{ kN}.$$

可直接得到剪力图, 如图所示.



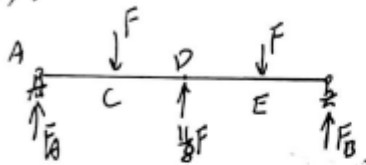
M图 (kN.m)

$Q_{AB} = 2 \text{ kN}$, $Q_{BA} = -14 \text{ kN}$, 跨中无集中力, 故剪力图为一条直线.



Q图 (kN)

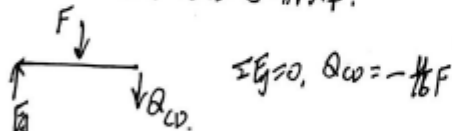
1f).



由左右对称性, 可得 $F_A = F_B = \frac{5}{16} F$.

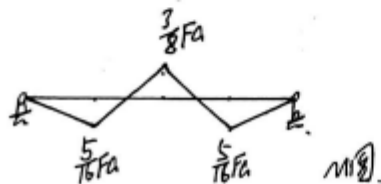
根据 $Q_{AC} = \frac{5}{16} F$, $M_A = 0$, 得 $M_{CA} = \frac{5}{16} F a$ (下侧受拉).

由 CD 中间截开并取左侧隔离体.

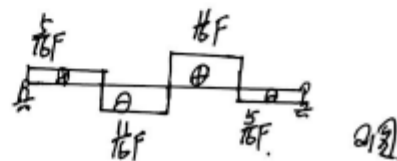


$$\sum F_y = 0, Q_{CD} = -\frac{11}{16} F$$

根据 CD 段剪力可得 $M_{DC} = \frac{3}{8} F a$ (上侧受拉), 则



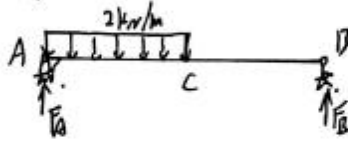
M图



Q图



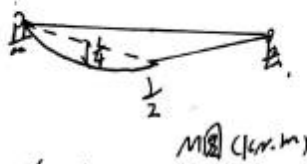
(9).



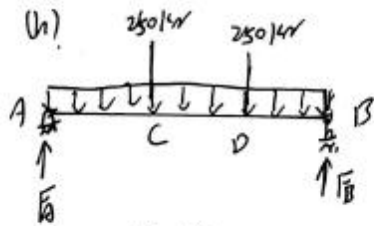
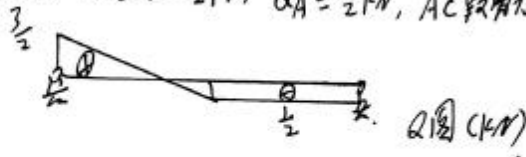
取整体隔离体, 由 $\sum M_A = 0$, $F_B \times 2 = 2 \times 1 \times \frac{1}{2}$, $F_B = \frac{1}{2} \text{ kN}$.

$\sum F_y = 0$, $F_A + F_B = 2 \times 1$, $F_A = \frac{3}{2} \text{ kN}$.

根据 $Q_{BC} = -\frac{1}{2} \text{ kN}$, $M_B = 0$, 可得 $M_{CB} = \frac{1}{2} \text{ kN}\cdot\text{m}$ (下侧受拉), 则弯矩图为

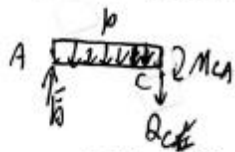


BC段剪力为 $Q_{BC} = -\frac{1}{2} \text{ kN}$, $Q_A = \frac{3}{2} \text{ kN}$, AC段剪力图斜率为 2 kN/m , 则剪力图为



由左右对称, 可得 $F_A = F_B = 280 \text{ kN}$.

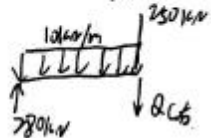
取 C 左侧到 A 点段隔离体



$\sum M_C = 0$, $M_{CA} + F_A \times 2 - 10 \times 2 \times 1 = 0$, $M_{CA} = -540 \text{ kN}\cdot\text{m}$.

$\sum F_y = 0$, $Q_{CA} + 10 \times 2 - F_A = 0$, $Q_{CA} = 260 \text{ kN}$.

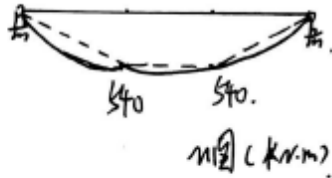
取 C 右侧到 A 点段隔离体



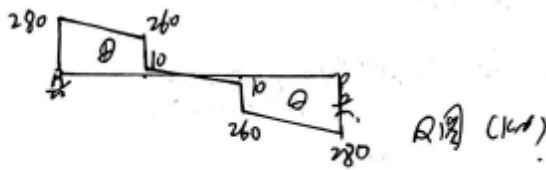
$\sum F_y = 0$, $Q_{CB} = 10 \text{ kN}$.



弯矩图为



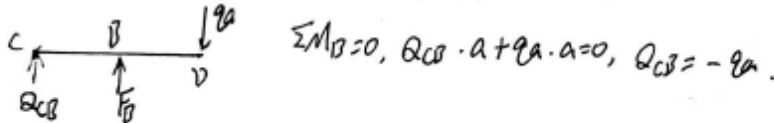
结构在荷载下, 剪力图为反弯矩。



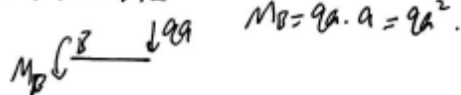
三、

(a).

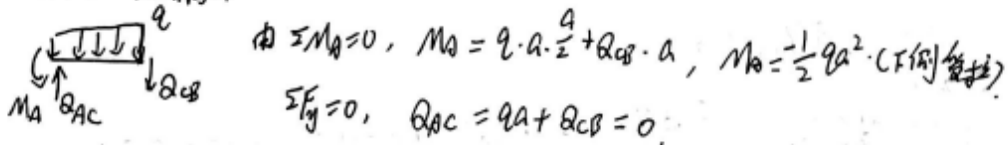
取BC段隔离体, 对B点取矩, 可得。



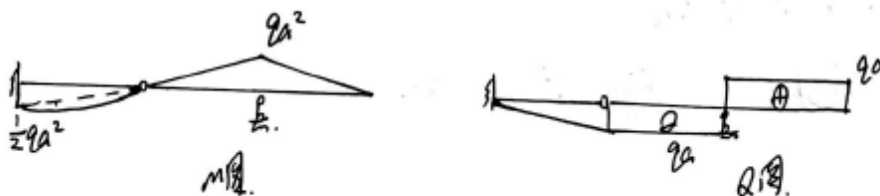
由BD段, 对B点取矩



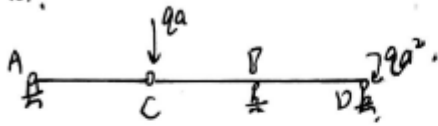
取AC段隔离体



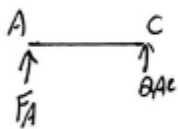
弯矩图、剪力图为



b).

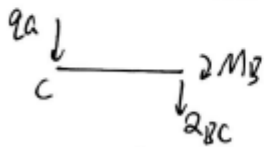


取AC段隔离体



由 $\sum M_C = 0$, 得 $F_A = 0 = Q_{AC}$, 故AC段无剪力、弯矩。

取B点左侧的C点隔离体。



由 $\sum M_B = 0$, $M_B = q_a \cdot a = q_a^2$.

$\sum F_y = 0$, $Q_{BC} = -q_a$.

则弯矩图、剪力图如图所示。

